Riccardo Venturi

Applied AI & Mechanical Engineer
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Portfolio: riccardo-venturi.github.io | github.com/Riccardo-Venturi

PROFESSIONAL SUMMARY

Mechanical Engineering graduate specializing in the intersection of Industry 4.0 and intelligent manufacturing systems. Proven ability to architect and deploy end-to-end MLOps pipelines (Docker, PyTorch) bridging probabilistic Deep Learning models with deterministic metrological measurements. Strong background in integrating complex datasets to solve physical engineering challenges (e.g., NDT composite inspection), with a focus on automation and reproducible infrastructure.

TECHNICAL SKILLS

AI & Computer Vision: Python, PyTorch, YOLOv8, UNet++, LSTM (TimeSeries), OpenCV.
Software & MLOps: Matlab, Linux, Windows, Docker, Docker-Compose, Git, PostgreSQL, CVAT, REST API.
Mechanical Engineering: 3D CAD Modeling (SolidWorks/Onshape), BricsCAD, Metrology, Technical Drafting.
Hardware & Embedded: ESP32, Arduino (C++), UART, I2C, Sensor Integration (IMU, Load Cells).
General Engineering: Operations Management, Cost-Benefit Analysis (NPV, ROI, ROR), Plant Layout Basics, Technical Documentation.

EXPERIENCE & APPLIED RESEARCH

AI/ML Developer | MLOps Engineer (Experimental Thesis) *Mar 2025 – Feb 2026*
University of Cagliari

- **Designed and deployed ROIA:** An industrial end-to-end Deep Learning pipeline for automated quantification of drilling-induced defects in CFRP radiographic NDT scans.
- **Computer Vision:** Architected a hierarchical pipeline (YOLOv8 + UNet++) using a custom hybrid *foveal* sampling strategy to capture low-contrast, sub-surface fibrous delaminations.
- **Metrology & ML:** Engineered a calibration workflow converting pixel-space segmentation into reliable mm^2 physical units using Random Forest regression, drastically improving precision ($R^2 = 0.061 \to 0.558$).
- **Predictive Maintenance:** Trained LSTM recurrent neural networks using Quantile Loss on GPU cloud instances to forecast tool wear evolution based on mechanical process data.
- **MLOps Deployment:** Containerized a full 19-service offline annotation infrastructure (CVAT, Traefik, PostgreSQL) via Docker, ensuring data sovereignty and deployment reproducibility.

TECHNICAL PROJECTS

Advanced Parametric CAD & Digital Prototyping *2021–2026*
Selected Projects (Details on my Digital Portfolio)

- **Structural & DFM Optimization:** Engineered industrial components (ES4 Manifold, ES7 Bracket) focusing on ribbing layouts, multi-axis intersections, and rigorous 2D drafting with tolerances.
- **Parametric Surfacing:** Managed robust feature trees and G2 curvature continuity (Project 6.1b) to prevent topological tearing during design iterations.
- **Linux & DevOps Workflow:** Integrated mechanical CAD with ray-traced rendering (CFL Bulb) to create high-fidelity digital twins. Managed project site deployment and automated asset optimization via **Ubuntu Terminal** and Git.

ESP32-Nano Bridge (Embedded Firmware) *2024*

EDUCATION & ADDITIONAL INFO

B.Sc. in Mechanical Engineering | University of Cagliari *2026*
Languages: Italian (Native), English: C1 Level (IELTS B2 certified)
Other: Driving License (B) | Relocation: Prague, DACH region, Italy.

ROIA — Automated CFRP Defect Quantification Pipeline

End-to-end computer vision and predictive maintenance for radiographic NDT inspection

Riccardo Venturi | riccardo.venturi.ing@gmail.com | github.com/Riccardo-Venturi

The Problem & Data Constraints: Manual radiograph inspection of drilling-induced delamination lacks metrological consistency. The project was constrained by a small dataset of 600+ industrial radiographic samples, characterized by **limited inter-hole variability** but **high morphological complexity** in the delamination patterns.

The Solution: A computer vision and MLOps pipeline combining semantic segmentation, morphological calibration, and temporal forecasting. Trained and Optimized for constrained GPU memory through hierarchical ROI extraction on cloud GPU infrastructure (Google Colab); deployable offline via local Docker stack with full data sovereignty.

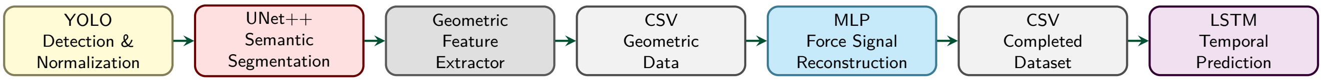


Figure 1: ROIA pipeline: YOLO → UNet++ → Feature Extraction → MLP → LSTM.

Key Technical Implementations

Key Contributions

- Designed a hierarchical CV pipeline (YOLOv8 + UNet++) bypassing full-scale hardware memory limits.
- Developed a hybrid foveal sampling strategy to target fibrous, low-contrast "spider-web" defect edges.
- Engineered a metrological calibration workflow mapping neural pixel clusters to reliable physical units (mm^2).
- Deployed a Cloud annotation infrastructure integrating CVAT, PostgreSQL, and Traefik via Docker.
- Designed a temporal predictive model using LSTM-based quantile regression.

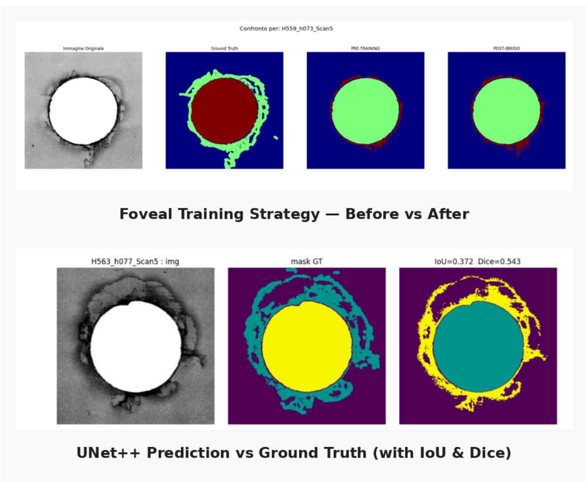


Figure 2: Semantic segmentation masks isolating defect geometry.

Quantitative Metrology Results

The core contribution transforms probabilistic segmentation masks into physically calibrated area estimates under limited-data conditions.

Pipeline Stage	Calibration Features	R^2 Score
1. Raw Segmentation	Neural pixel area	0.061
2. Geometric Calibration	Morphology (RF)	0.489
3. Temporal Modeling	+ Wear-sequence (LSTM)	**0.558**

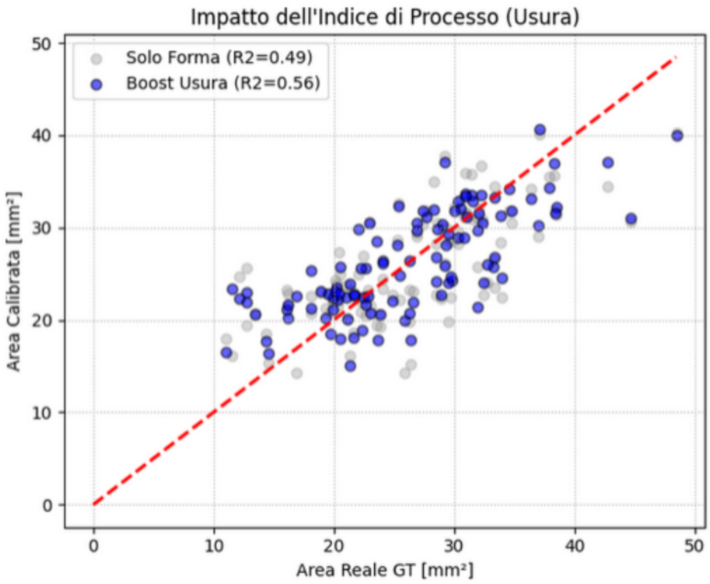


Figure 3: Error compensation via morphological calibration.

Constraints & Industrial Tech Stack

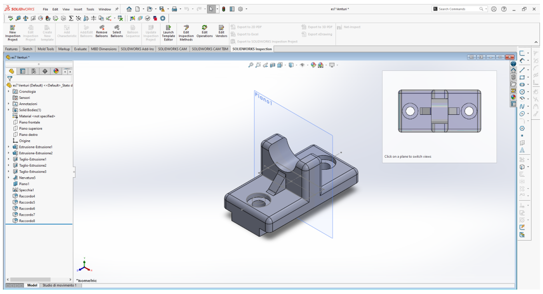
Failure Modes (Deployment limits):

- **Annotation Noise:** Inherently ambiguous edge definition in CFRP restricts the absolute accuracy ceiling.
- **Edge Constraints:** Optimized for offline batch processing; requires weight quantization for real-time edge CNC integration.

Core ML: PyTorch, YOLOv8, UNet++, LSTM
Metrology: Scikit-learn (Random Forest), OpenCV
Infrastructure: Docker, Docker-Compose, CVAT
Environment: Linux (Ubuntu), Git

MECHANICAL ENGINEERING PORTFOLIO

PROJECT 01: INDUSTRIAL SUPPORT BRACKET (ES7)



Technical Depth: High-rigidity support designed for cylindrical housing.

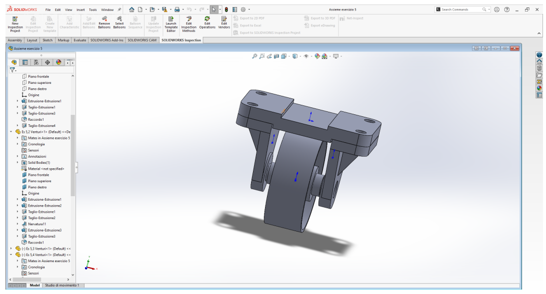
- **Features:** Integrated reinforcement ribs, dual-bolt mounting pattern, R5 safety fillets.
- **ISO Standards:** Geometric dimensioning and tolerancing (GD&T) according to ISO 2768-m.
- **Modeling:** Advanced solid modeling with bilateral symmetry and section analysis.

See full A2 Technical Drawing in Appendix.

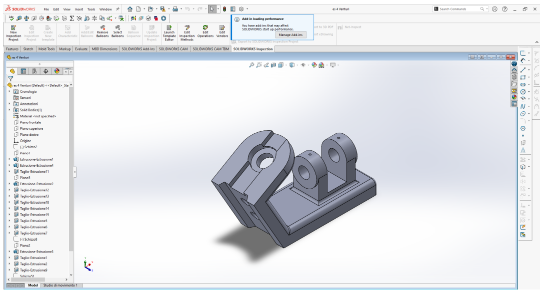
PROJECT 02: KINEMATIC WHEEL SUB-ASSEMBLY (ES5)

Focus: Mechanical assembly and motion constraints.

- **Complexity:** 12+ unique components managed via robust Mates.
- **Engineering:** Dynamic load distribution study between pivot housing and wheel axle.
- **Documentation:** Exploded views and Bill of Materials (BOM) management.



PROJECT 03: MANIFOLD MOUNTING ADAPTER (ES4)



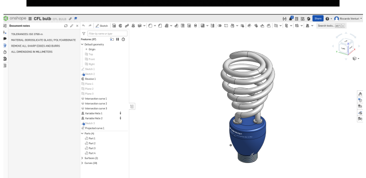
Design for Manufacturing (DFM):

- **Geometry:** Multi-axis hole intersections with specific draft angles for casting.
- **Optimization:** Strategic material removal to reduce weight while maintaining structural integrity.
- **Drafting:** 2D detailed layout with localized section views.

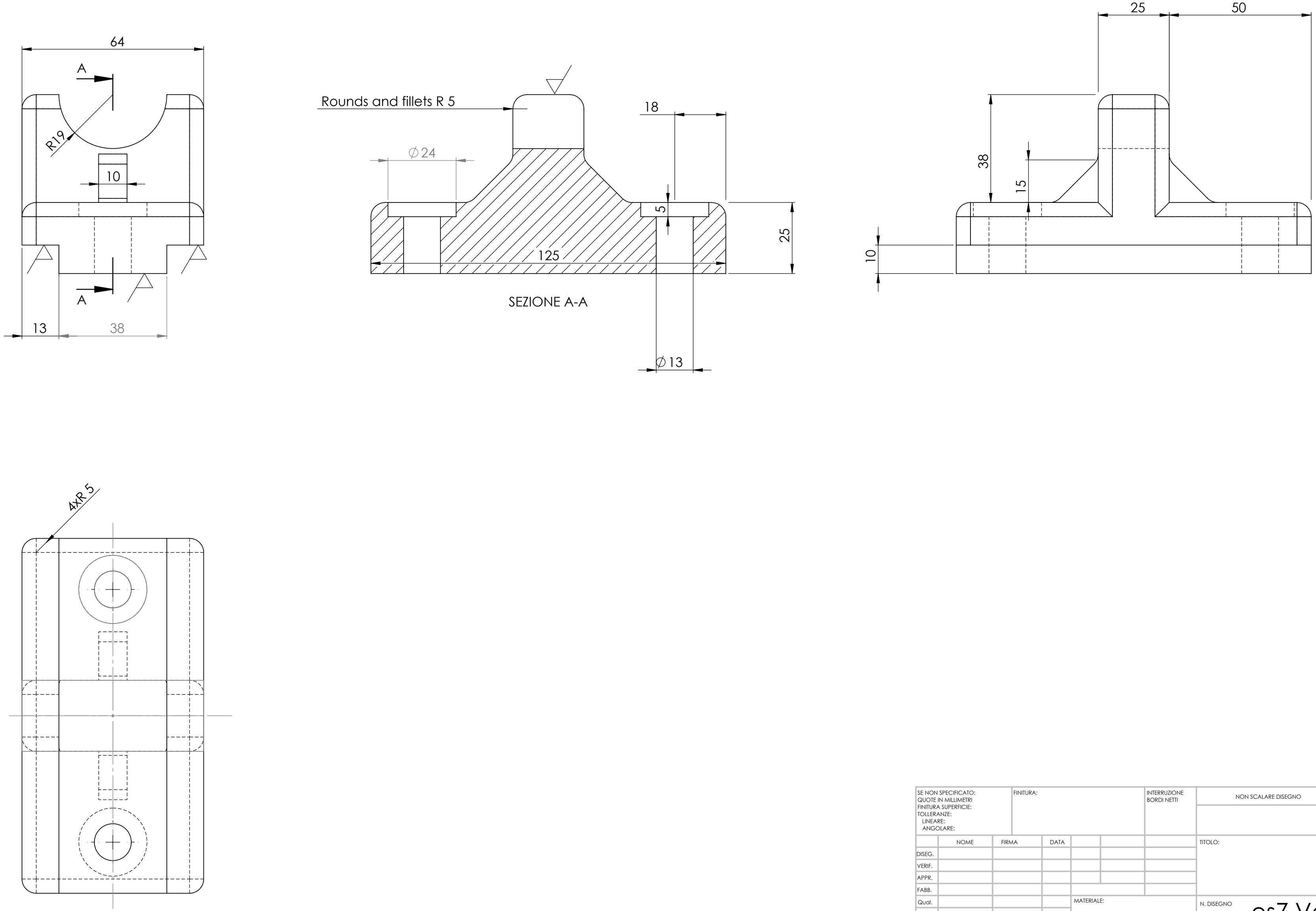
PROJECT ONSHAPE N.1: CFL BULB LIGHT

Focus: Splines and Curves on CAD Software.

- **Complexity:** Fotorealistic Render in Blender followed CAD creation.
- **Engineering:** Work on a lightbulb with complex Geometry and glass render.
- **Documentation:** Render in Blender and Worktree on Onshape.



APPENDIX: Original A2/A3 technical sheets with title blocks are attached. Interactive 3D models: **riccardo-venturi.github.io**



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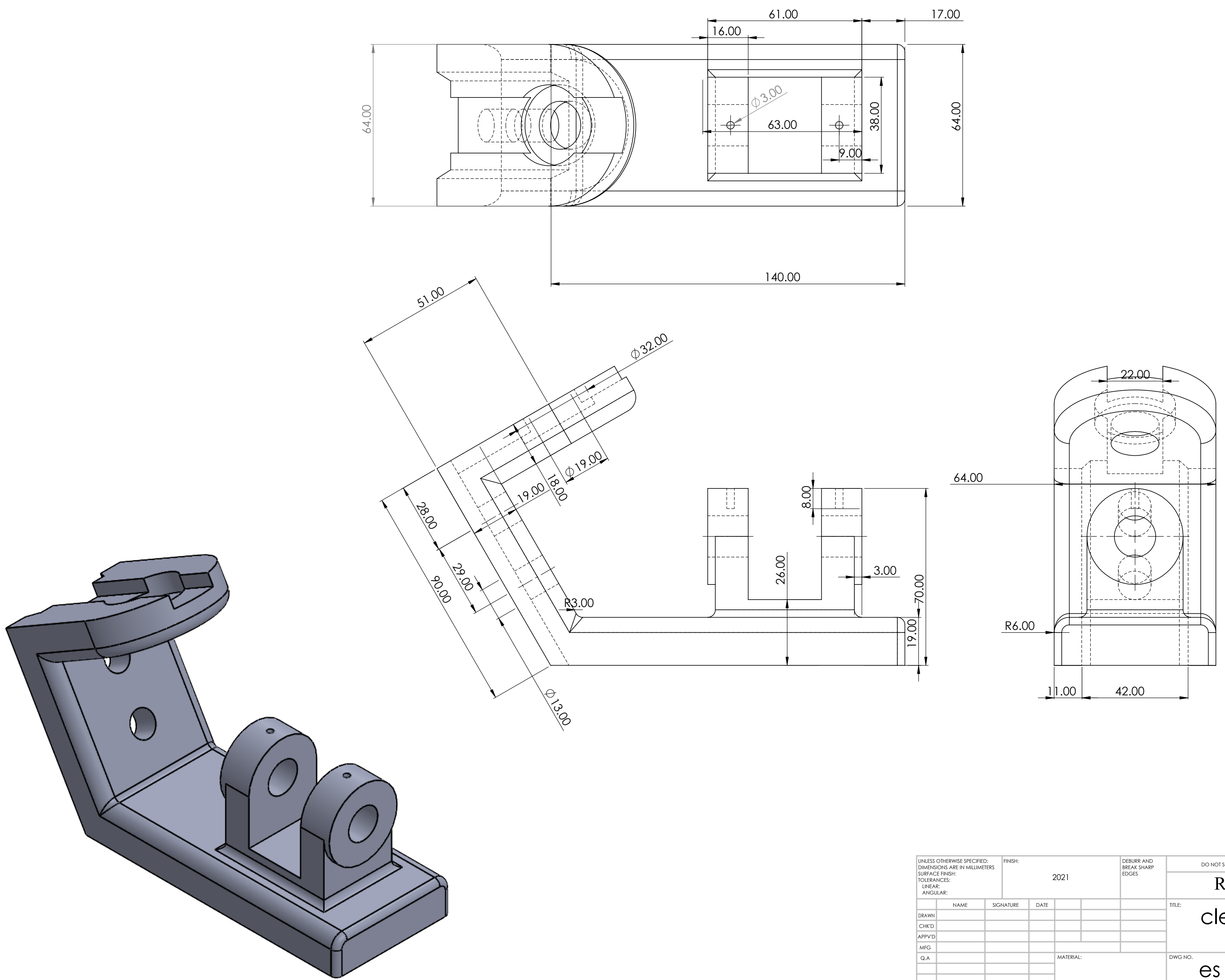

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2021

DEBURR AND BREAK SHARP EDGES

DO NOT SCALE DRAWING

REVISION

Riccardo Venturi

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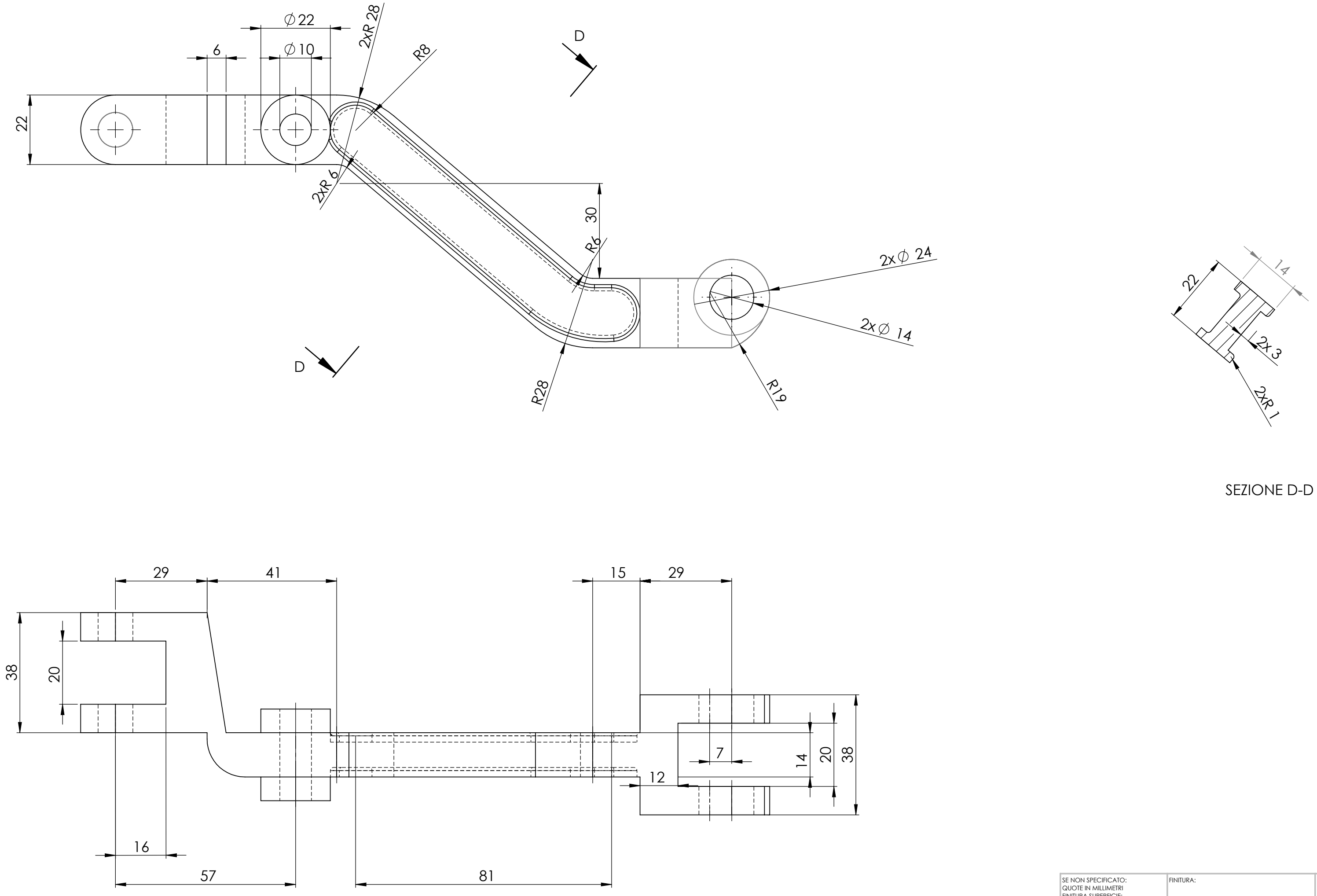

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